```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, StratifiedShuffleSplit
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.base import BaseEstimator, TransformerMixin
```

```
url = "https://media.geeksforgeeks.org/wp-content/uploads/20240319120216/housing.csv"
housing = pd.read_csv(url)

housing.head()
```

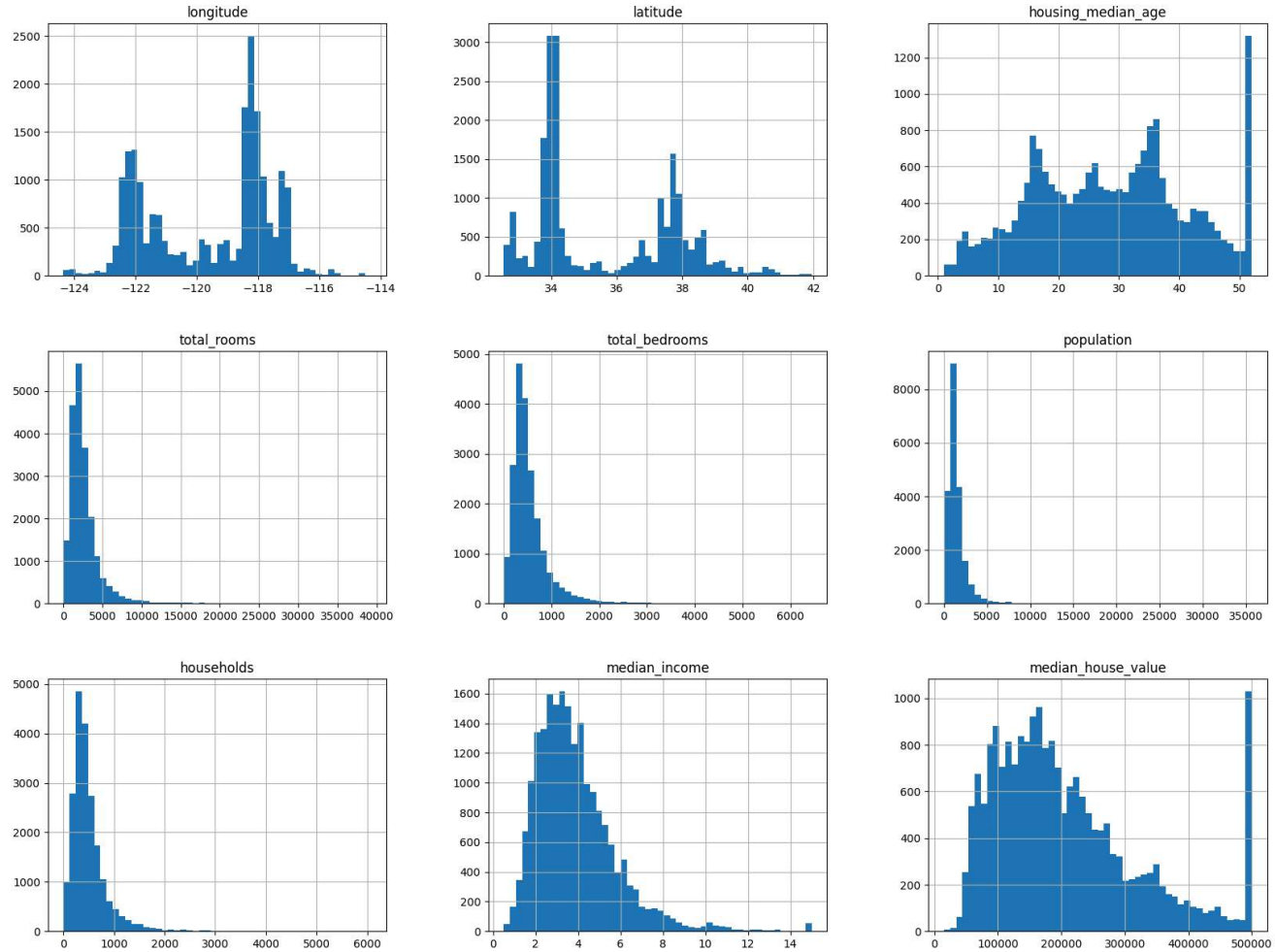
	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_valu
0	-122.23	37.88	41.0	880.0	129.0	322.0	126.0	8.3252	452600
1	-122.22	37.86	21.0	7099.0	1106.0	2401.0	1138.0	8.3014	358500
2	-122.24	37.85	52.0	1467.0	190.0	496.0	177.0	7.2574	352100
3	-122.25	37.85	52.0	1274.0	235.0	558.0	219.0	5.6431	341300
4	-122.25	37.85	52.0	1627.0	280.0	565.0	259.0	3.8462	342200

```
housing.info()
housing.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 10 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   longitude           20640 non-null  float64
 1   latitude            20640 non-null  float64
 2   housing_median_age  20640 non-null  float64
 3   total_rooms         20640 non-null  float64
 4   total_bedrooms      20433 non-null  float64
 5   population          20640 non-null  float64
 6   households          20640 non-null  float64
 7   median_income       20640 non-null  float64
 8   median_house_value  20640 non-null  float64
 9   ocean_proximity     20640 non-null  object
dtypes: float64(9), object(1)
memory usage: 1.6+ MB
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	med
count	20640.000000	20640.000000	20640.000000	20640.000000	20433.000000	20640.000000	20640.000000	20640.000000	
mean	-119.569704	35.631861	28.639486	2635.763081	537.870553	1425.476744	499.539680	3.870671	
std	2.003532	2.135952	12.585558	2181.615252	421.385070	1132.462122	382.329753	1.899822	
min	-124.350000	32.540000	1.000000	2.000000	1.000000	3.000000	1.000000	0.499900	
25%	-121.800000	33.930000	18.000000	1447.750000	296.000000	787.000000	280.000000	2.563400	
50%	-118.490000	34.260000	29.000000	2127.000000	435.000000	1166.000000	409.000000	3.534800	
75%	-118.010000	37.710000	37.000000	3148.000000	647.000000	1725.000000	605.000000	4.743250	
max	-114.310000	41.950000	52.000000	39320.000000	6445.000000	35682.000000	6082.000000	15.000100	

```
housing.hist(bins=50, figsize=(20,15))
plt.show()
```

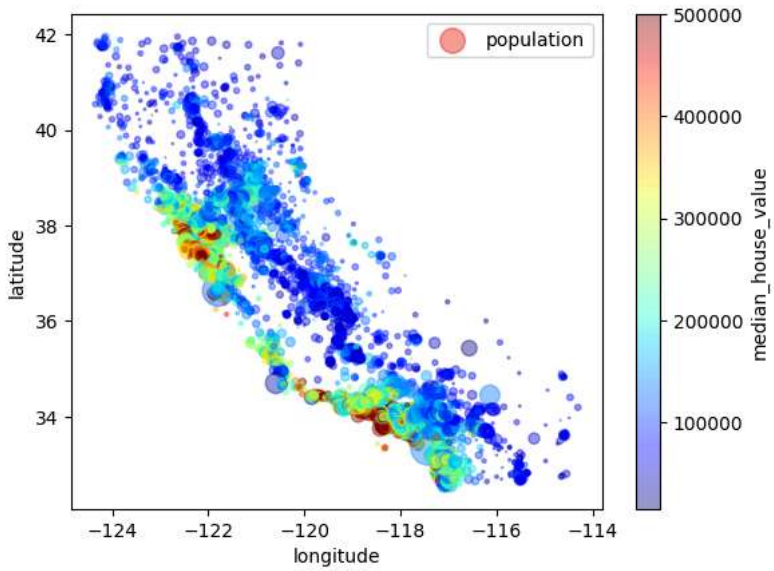



```
train_set, test_set = train_test_split(housing, test_size=0.2, random_state=42)
```

```
housing["income_cat"] = pd.cut(housing["median_income"],
                               bins=[0.,1.5,3.0,4.5,6.,np.inf],
                               labels=[1,2,3,4,5])

split = StratifiedShuffleSplit(n_splits=1, test_size=0.2, random_state=42)

for train_index, test_index in split.split(housing, housing["income_cat"]):
    strat_train_set = housing.loc[train_index]
    strat_test_set = housing.loc[test_index]
```

```
housing.plot(kind="scatter", x="longitude", y="latitude",
             alpha=0.4,
             s=housing["population"]/100,
             label="population",
```

```
             c="median_house_value",
             cmap=plt.get_cmap("jet"),
             colorbar=True)

plt.legend()
plt.show()
```

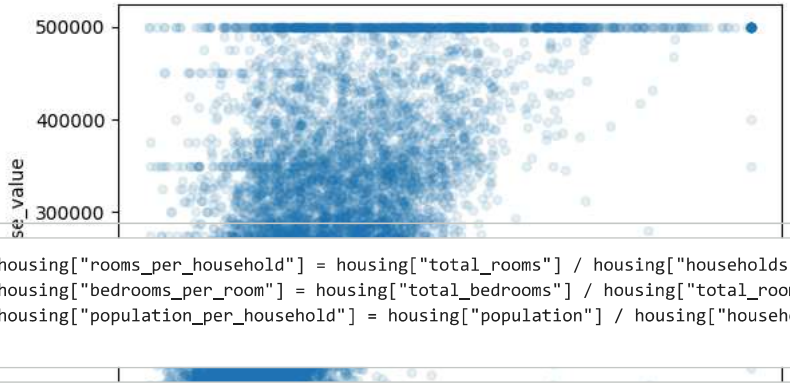



```
corr_matrix = housing.corr(numeric_only=True)
corr_matrix["median_house_value"].sort_values(ascending=False)
```

	median_house_value
median_house_value	1.000000
median_income	0.688075
total_rooms	0.134153
housing_median_age	0.105623
households	0.065843
total_bedrooms	0.049686
population	-0.024650
longitude	-0.045967
latitude	-0.144160

dtype: float64

```
housing.plot(kind="scatter",
             x="median_income",
             y="median_house_value",
             alpha=0.1)
plt.show()
```




```
housing["rooms_per_household"] = housing["total_rooms"] / housing["households"]
housing["bedrooms_per_room"] = housing["total_bedrooms"] / housing["total_rooms"]
housing["population_per_household"] = housing["population"] / housing["households"]
```

```
corr_matrix = housing.corr(numeric_only=True)
corr_matrix["median_house_value"].sort_values(ascending=False)
```

median_house_value	1.000000
median_income	0.688075
rooms_per_household	0.151948
total_rooms	0.134153
housing_median_age	0.105623
households	0.065843
total_bedrooms	0.049686
population_per_household	-0.023737
population	-0.024650
longitude	-0.045967
latitude	-0.144160
bedrooms_per_room	-0.255880

dtype: float64

```
from sklearn.impute import SimpleImputer

imputer = SimpleImputer(strategy="median")
housing_num = housing.drop("ocean_proximity", axis=1)
imputer.fit(housing_num)

housing_num_imputed = imputer.transform(housing_num)
```

```
housing_cat = housing[["ocean_proximity"]]

encoder = OneHotEncoder()
housing_cat_1hot = encoder.fit_transform(housing_cat)

housing_cat_1hot.toarray()
```

```
array([[0., 0., 0., 1., 0.],
       [0., 0., 0., 1., 0.],
       [0., 0., 0., 1., 0.],
       ...,
       [0., 1., 0., 0., 0.],
       [0., 1., 0., 0., 0.],
       [0., 1., 0., 0., 0.]])
```

```
scaler = StandardScaler()
housing_scaled = scaler.fit_transform(housing_num_imputed)
```

```
class CombinedAttributesAdder(BaseEstimator, TransformerMixin):
    def fit(self, X, y=None):
        return self

    def transform(self, X):
        rooms_per_household = X[:,3] / X[:,6]
        population_per_household = X[:,5] / X[:,6]
        bedrooms_per_room = X[:,4] / X[:,3]

        return np.c_[X,
                     rooms_per_household,
                     population_per_household,
                     bedrooms_per_room]
```

```
housing = strat_train_set.drop("median_house_value", axis=1)
housing_labels = strat_train_set["median_house_value"]

num_attribs = list(housing.drop("ocean_proximity", axis=1))
cat_attribs = ["ocean_proximity"]
```

```
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer

num_pipeline = Pipeline([
        ('imputer', SimpleImputer(strategy="median")),
        ('attribs_adder', CombinedAttributesAdder()),
        ('scaler', StandardScaler()),
    ])

full_pipeline = ColumnTransformer([
        ("num", num_pipeline, num_attribs),
        ("cat", OneHotEncoder(), cat_attribs),
    ])

housing_prepared = full_pipeline.fit_transform(housing)
```

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(housing_prepared, housing_labels)
```

```
▾ LinearRegression
LinearRegression()
```